



MASC: Joint motion-aware and VLM-semantic augmentation for contextual traffic anomaly detection

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ABSTRACT

Traffic anomaly detection in driving videos is crucial for ensuring the safety and reliability of autonomous driving systems. However, existing methods often focus on pixel-level scene dynamics or object behaviors, failing to comprehensively model multi-level motion patterns and high-level semantic relationships. As a result, models tend to be insensitive to abnormal events that exhibit subtle pixel changes but violate normal semantic relationships. To achieve unified modeling of motion and semantics, we propose MASC, a cross-modal semantic-enhanced framework for traffic anomaly detection. The proposed method introduces semantic embeddings generated by a vision language model as priors, elevating driving video representations from pixel-level motion patterns to a semantic-level space. Through a semantic-enhanced motion alignment mechanism, the framework learns high-level representations with improved interpretability. Furthermore, to identify abnormal semantic relationships, MASC incorporates a relational reasoning module that captures the interactions among traffic participants, behavioral patterns, and environmental constraints, thereby enabling context-aware anomaly detection at both scene and instance levels. Experiments demonstrate that MASC achieves competitive performance in terms of detection accuracy and generalization with over 1.0% AUC improvement on both the DoTA and DADA datasets. Further analysis indicates that incorporating semantic priors not only enhances interpretability but also improves early identification of anomalies by strengthening the modeling of interactive relationships in scenarios.

1. Introduction

Traffic anomaly detection (TAD) plays a key role in autonomous driving and intelligent transportation systems. Its goal is to identify potential risks in complex and continuously changing road environments, including vehicle loss of control, pedestrian intrusion, traffic collisions, and unexpected obstacles [1]. Accurate and timely detection of such anomalies enhances an autonomous system's ability to understand hazardous situations and provides crucial support for early risk warning, emergency decision-making, and accident prevention [2]. Therefore, developing a robust, reliable, and generalizable framework for traffic anomaly detection is of great importance for improving transportation safety.

Real-world traffic scenes exhibit strong temporal variability, as the

motion states of various road participants and surrounding environmental conditions change continuously. Moreover, traffic environments contain rich semantic elements whose attributes and roles are highly complex, such as structural layouts, functional regions, and interaction constraints [1,3]. These factors collectively make high-quality anomaly detection in real driving environments a challenging task. Existing traffic anomaly detection approaches can be broadly categorized into two major branches: scene-level methods and instance-level methods. Scene-level methods focus on learning global visual features or video dynamics to detect anomalies [3,4]. While these approaches are effective in capturing overall environmental changes, they often struggle to detect critical anomalous cues triggered by individual road users, such as a pedestrian suddenly crossing the road or a vehicle rapidly changing lane. In contrast, instance-level methods concentrate on the local

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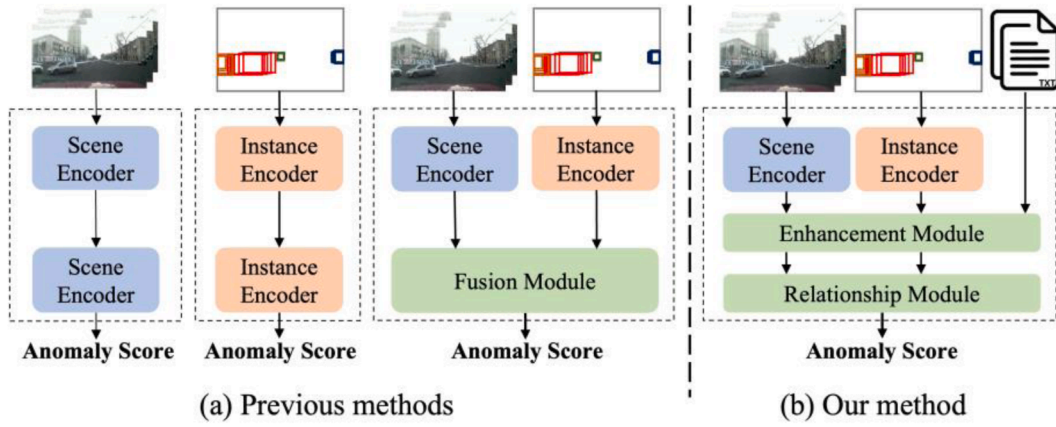


Fig. 1. Previous TAD frameworks (a) vs. the proposed framework (b).

behavior of individual objects, identifying anomalies through detection boxes, motion trajectories, or short-term behavior fluctuations [5,6]. Although these methods are better at modeling object dynamics, they lack the contextual semantic constraints required to determine anomalies dependent on scene semantics, such as lane departure, wrong-way driving, etc. In addition, instance-level methods are highly susceptible to errors from the object detection stage and often struggle to handle scenes without explicit object instances.

In recent years, several studies [7–9] have attempted to jointly model scene-level and instance-level features to overcome the representational limitations of single-branch architectures in complex traffic scenarios. Although these methods integrate global scene dynamics with local object motion, their fusion strategies essentially remain at the pixel-level or feature-level alignment stage, as illustrated in Fig. 1(a). Specifically, the scene branch and the instance branch are combined through simple concatenation, weighting, or attention mechanisms, lacking a unified modeling framework at the semantic level. In practice, scene motion is often influenced by ego-motion and is deeply coupled with the behavioral changes of traffic participants [10]. For example, the synchronized displacement between vehicles and road structures, or the coordinated evolution of trajectories among multiple traffic participants, fundamentally depends on semantic constraints among roles, relationships, and environmental context, rather than merely spatiotemporal correspondences in pixel space. Due to the absence of semantic modeling, existing multi-branch detectors primarily rely on local pixel deviations, optical flow discrepancies, or appearance anomalies as detection cues, making it difficult to understand the structured relationships between scene evolution and individual behaviors from a unified perspective. Although some studies [11,12] attempt to enhance semantic supervision by introducing fine-grained textual descriptions, such approaches depend on costly and labor-intensive annotations, which limit their scalability in real-world applications.

To address the challenges above, we introduce Motion-Aware and VLM-Semantic Augmentation for Contextual Traffic Anomaly Detection (MASC), a unified framework for understanding contextual anomalies in traffic scenes, as shown in Fig. 1(b). The core idea of MASC is that traffic anomalies are not triggered by a single factor; rather, they arise from the joint influence of scene structures, participant behaviors, semantic roles, and dynamic interactions. Motivated by this observation, MASC reconstructs the feature representation space from three complementary dimensions to achieve a comprehensive understanding of complex driving environments, including scene-level context, instance-level dynamics, and semantic priors. The rapid advancement of vision-language models has significantly improved long-range dependency modeling and contextual reasoning, enabling more robust, interpretable, and scalable solutions for video anomaly detection [13]. Moreover, recent studies [14,15] have shown that VLMs possess strong capabilities in traffic

scene understanding, including object recognition, spatial localization, and holistic scene interpretation. Based on this, MASC incorporates coarse semantic priors derived from a generic VLM to enhance both the scene and instance branches. This process enables the construction of scene-level and instance-level representations enriched with diverse attributes, roles, and interaction cues, while also promoting semantic consistency in traffic anomaly understanding [16]. To further integrate semantic signals across objects and structural elements, we introduce a unified semantic relation reasoning module. Through a set of learnable query tokens, this module jointly attends to scene semantics, object semantics, motion cues, and visual structures, thereby forming a global semantic relation space that reflects anomalies in traffic scenarios. Finally, MASC employs a dual-branch decoder that predicts scene-level and instance-level anomaly scores, respectively, enabling anomaly reasoning from fine-grained object behavior to a holistic scene.

The main contributions of our work can be summarized as follows:

1. We propose a cross-modal semantic enhancement mechanism that elevates driving video representations from pixel-level patterns to a semantic-level space. MASC leverages text embeddings generated by a Vision-Language Model (VLM) as semantic priors and aligns motion features with semantic information, enabling the model to learn high-level representations with semantic interpretability.
2. We perform anomaly detection based on multi-branch semantic relationships. Unlike approaches that rely on pixel variations or optical flow deviations, MASC models the semantic relationships among traffic roles, behavioral patterns, and environmental constraints. Through a relational reasoning module, it captures contextual dependencies among instances and identifies anomalous events that violate semantic conventions or interaction logic.
3. The proposed method delivers strong performance on widely used traffic anomaly detection benchmarks (DoTA and DADA), confirming the effectiveness of combining motion-aware modeling with semantic augmentation. Moreover, MASC demonstrates that VLM-generated coarse semantic embeddings can serve as effective feature-level priors rather than supervision signals, providing a practical reference for video understanding scenarios where fine-grained textual labels are scarce or unavailable.

2. Related work

2.1. Video anomaly detection

Traffic Anomaly Detection (TAD) largely originates from the classical field of Video Anomaly Detection (VAD). Early VAD studies primarily focused on identifying anomalous events under fixed surveillance viewpoints and were widely applied to diverse scenarios such as public

safety, social behavior, and traffic incidents [17]. With the evolution of deep learning techniques, the field of video anomaly detection has gradually developed a motion-centric technical paradigm in recent years. Under this paradigm, research typically aims to learn spatio-temporal dynamic patterns in normal scenes, defining anomalous events as significant deviations from established motion distributions. Specifically, mainstream approaches include flow-based modeling [18, 19], temporal gradient learning [3, 20], and motion decoupling [16, 21], all of which aim to accurately evaluate the rarity and irregularity of motion features. For example, Wang et al. [16] enhanced semantic coordination among multi-dimensional features of complex anomaly by integrating a three-branch encoder for appearance, motion, and pose with a spatial channel complementarity module. Lyu et al. [21] reformulated anomaly detection as a Gaussian deblurring process and incorporated a background-agnostic motion memory module to decouple spatial and motion features, thereby effectively mitigating the over-generalization of generative models to anomalous samples. However, although these motion-centered methods perform well in constrained viewpoint scenarios, their generalization ability is often significantly limited when confronted with drastic viewpoint shifts or complex dynamic background interference.

2.2. Traffic anomaly detection

Early research on traffic anomaly detection in driving videos inherited the core ideas of traditional VAD, namely identifying anomalous frames by modeling global temporal dynamics [22, 23]. Rossi et al. [24] construct an online anomaly detection framework with a short-term module extracting spatiotemporal features via Video Swin Transformer and a long-term module that accumulates temporal context. Zhou et al. [25] propose a coarse-to-fine framework that first detects potential accident frames through HOF-based temporal encoding and then applies Faster R-CNN to extract object features and spatial relations for refined detection. These methods focus on learning scene-level motion patterns and treat anomalies as deviations from the overall spatio-temporal distribution. A few studies adopt purely instance-level modeling strategies, identifying anomalous objects by analyzing temporal variations in detected bounding boxes or predicted trajectories [26, 27]. For example, Yao et al. [7] predicted trajectory consistency and inferred whether an object was involved in an accident by jointly modeling object motion and ego-motion, where ego-motion is obtained from the trajectory planning of the autonomous vehicle. Unlike traditional VAD methods that rely on static surveillance viewpoints, driving videos exhibit continuously changing perspectives due to ego-vehicle motion. This results in more complex scene dynamics, diverse behaviors of traffic participants, and substantial environmental noise. Under such circumstances, modeling based on a single motion source often struggles to distinguish between background motion and anomalous behavior.

To alleviate the aforementioned issues, researchers have attempted to integrate multi-dimensional feature information. Some studies incorporate prior knowledge such as object information [9, 28], driver's attention [4], and scene descriptions [11, 12] into a single motion branch to enhance the model's ability for focusing on key areas. For example, Qiu et al. [9] proposed the PromptTAD framework, which achieves scene-object relationship modeling under frame-level and object-level supervision. It introduces instance and relationship cross-attention modules, utilizing static object detection bounding boxes as prompts to extract anomalous clues from temporal features within frames. Li et al. [4] reconstruct driver attention through dense semantic descriptions, enabling models to learn human cognitive patterns for recognizing traffic anomalies. Although these methods introduce richer prior guidance, they still rely on single-source motion modeling, overlooking the coupling relationship between background motion and object motion, which may lead to inaccurate anomaly localization. Another research direction focuses on joint motion modeling. These

methods independently model background motion and object motion, and adopt a parallel optimization framework consisting of a "robustness task" and a "precision task" to alleviate the conflict in a single objective function that struggles to balance accuracy and robustness [29, 30]. Yao et al. [31] propose the FOL-Ensemble, which combines future RGB prediction based on U-Net with future bounding box prediction based on optical flow and box trajectory, fusing both signals in pixel space for joint anomaly detection. Fang et al. [7] add a contextual consistency branch based on object-enhanced images to learn relational features for mitigating background interference. Liang et al. [8] introduce the MAMTCF framework, which integrates global motion modeling by optical-flow reconstruction and local motion modeling by future object localization, further enhanced by memory-augmented high-dimensional normal motion representations. However, such methods lack semantic consistency constraints between object behaviors and scene context. When anomalies manifest as semantic-level inconsistencies rather than significant pixel variations, these models often struggle to identify them effectively. To address these limitations, this paper introduces semantic prior information into the joint motion modeling framework to enhance semantic consistency between object-level and scene-level features.

2.2. Application of traffic scene semantics

In recent years, the rapid development of Vision Language Model (VLM) has significantly advanced the field of video understanding. With the maturation of cross-modal pretraining frameworks, VLM has gradually evolved from image-text alignment to semantic modeling at the video level, achieving remarkable performance improvements in tasks such as video captioning, event localization, visual question answering, and cross-modal retrieval [32, 33]. For example, Li et al. [34] developed VideoChat by integrating a spatio-temporal feature aggregator with a visual encoder to produce compact video representations, which are then embedded into a pretrained large language model. Through specialized fine-tuning datasets, they built a video dialogue model with precise spatio-temporal perception and interactive reasoning capabilities. Xu et al. [35] proposed the LLaVA-CoT framework, which introduces a structured four-stage reasoning paradigm and a stepwise backtracking search mechanism, significantly enhancing the systematic understanding and self-correction capabilities of vision language models in complex tasks, such as instance reasoning, logical reasoning, and math. Chen et al. [36] constructed an automated 3D spatial data synthesis pipeline based on multi-expert model collaboration, significantly improving the performance of vision language models in quantitative reasoning within 3D spaces.

Modern vision language models align visual features with natural language, enabling structured understanding of roles, behaviors, and context in dynamic scenes. Owing to the abstract and structured nature of language, they capture object attributes, intentions, and semantic relationships beyond pixel-level appearance and motion variations [37, 38]. Such high-level semantics offer new perspectives for behavior understanding and risk analysis in complex traffic scenarios. The first line of work adopts CLIP-style cross-modal contrastive learning to align textual semantics with visual representations, thereby enhancing the discriminative capability of downstream tasks. For example, Liang et al. [11] and Lohner et al. [12] map fine-grained textual descriptions of traffic anomalies into a high-dimensional semantic space and employ contrastive supervision to align language and vision features, providing richer and more comprehensive semantic guidance for anomaly detection. Similar strategies have also been explored in autonomous driving decision making [39], traffic scene understanding [40], and vehicle behavior prediction [41]. The second line of work directly uses text-generated semantics vector as auxiliary inputs while training models with the original task supervision. These approaches do not rely on contrastive pretraining but instead incorporate high-level information from text descriptions to enhance understanding of scene roles, spatial relationships, and behavioral intent. For instance, Kuang et al.

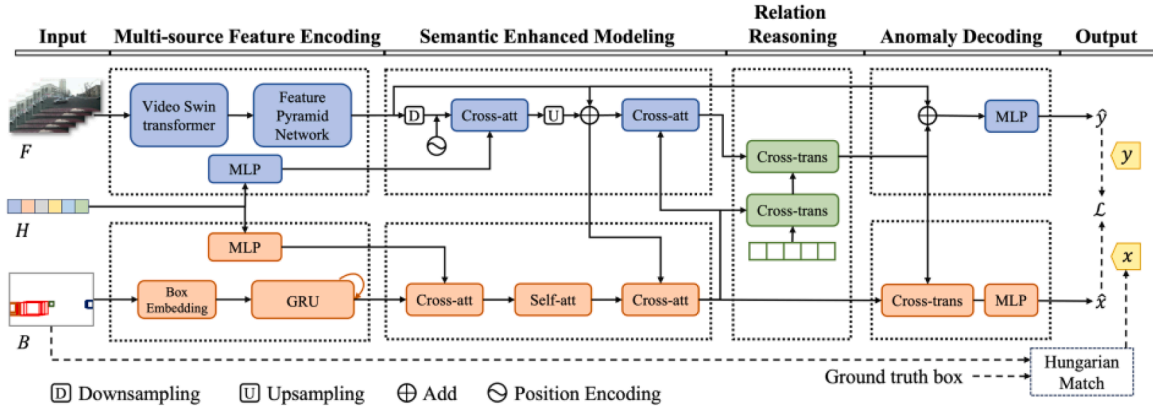


Fig. 2. The framework of the proposed method. The architecture consists of a scene branch and an instance branch. The scene branch, shown in the blue part of the framework, performs scene-level anomaly prediction and takes the RGB frames F of a video clip as its primary input. The instance branch, shown in the red part, conducts object-level anomaly detection using the detected bounding boxes B within the video clip. In addition, the semantic embeddings H generated by the VLM are injected into both branches as auxiliary inputs to enhance their semantic understanding.

[42] leverage VLMs to construct a comprehensive traffic knowledge graph and subsequently derive interpretable driving decisions. Zheng et al. [43] utilized structured and costly driving scenario information obtained from large language models to enhance the reliability of motion prediction. Li et al. [4] reconstruct driver attention through dense semantic descriptions, enabling models to learn human cognitive patterns for recognizing traffic anomalies.

Despite their successes, these methods heavily depend on manually crafted, fine-grained and structured textual annotations, which are costly, labor-intensive, and difficult to scale to open-world driving scenarios. Such reliance limits the end-to-end capability of the models and constrains the practicality of semantic enhancement in real autonomous driving systems. Motivated by these challenges, this work leverages automatically generated coarse-grained semantics from generic VLM as a label-free prior, enabling low-cost acquisition of high-level semantic cues to enhance contextual understanding in traffic anomaly detection.

3. The proposed method

In this section, we introduce the proposed MASC framework. Our approach is conceptually inspired by PromptTAD [9], which leverages detection-box prompts to guide information aggregation between object and scene features, as well as to model inter-object relationships for traffic anomaly detection. However, PromptTAD primarily focuses on appearance-based object cues, making it difficult to capture abnormal motion patterns and lacking higher-level semantic understanding of the traffic scene. To address these limitations, MASC adopts a scene-context enhancement strategy that injects instance-level motion representations and VLM-derived semantic priors into the model, thereby improving the detection of abnormal behaviors. Fig. 2 illustrates the overall architecture of our framework.

3.1. Data preparation and preprocessing

Fig. 2 illustrates the three types of inputs consumed by MASC for a video clip of length T , including the RGB frames, the object detection boxes, and the semantic embeddings. The RGB frames provide appearance information of the scene, and their temporal continuity reflects scene-level motion patterns, including both ego-motion and the motion of surrounding traffic participants. To obtain explicit spatial positions of these participants, we extract object bounding boxes from consecutive frames using a pretrained YOLOv9 detector [44]. While these two inputs provide clear cues regarding appearance changes and object motion, they lack higher-level semantic attributes, such as object categories, functional roles, scene context, and interaction relationships among traffic participants.

To address this limitation, we introduce semantics generated by a generic Vision-Language Model (VLM) to enrich the model's understanding of both the scene and the objects. We follow the principle of complex task decomposition by splitting scene understanding into two stages: object identification and object comprehension [30]. By providing object-level prompts, we guide the VLM to focus on the object comprehension stage, thereby obtaining high-quality semantic features [45]. As illustrated in Fig. 3, we take the detected bounding boxes from YOLOv9 as positional prompts to form an object-enhanced image and use specific instructions to direct the VLM's attention toward these regions. From the textual outputs, we observe that the VLM can describe the current traffic scene, identify categories of road users, and capture coarse relational attributes such as spatial interactions. Naturally, due to hallucination and noise, these descriptions can be coarse or partially incorrect.

However, directly using the VLM's textual output requires converting visual features into natural language and then mapping text back into a vector. This process introduces semantic discretization and re-embedding inconsistencies, which inevitably compress or distort the

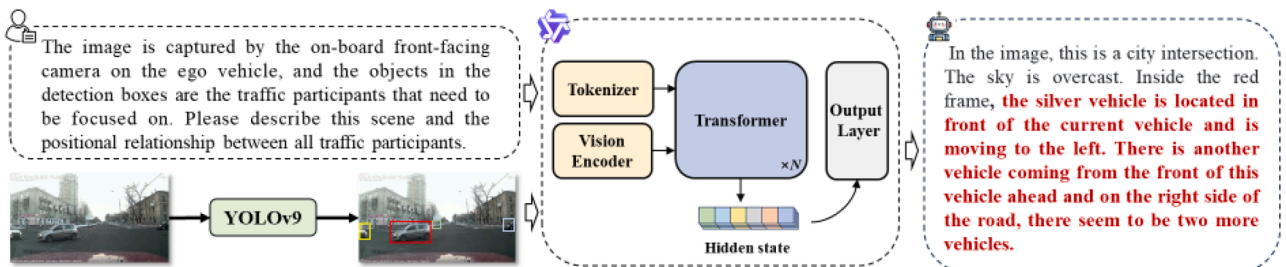


Fig. 3. The generation mechanism of semantic description.

fine-grained cross-modal semantics originally encoded within the VLM. Such a conversion not only reduces the reliability of the semantic priors but also amplifies hallucinations and incomplete descriptions. Therefore, we adopt the VLM's hidden-state representations H as continuous, high-dimensional semantic priors. This avoids the information loss and instability inherent in textual outputs while preserving richer and more stable cross-modal semantics for downstream fusion.

$$Z_t = \text{VLM}(f_1) = [z_1, z_2 \dots z_t] \in \mathbb{R}^{n \times D_s} \quad (1)$$

where $F_t = [f_1, f_2, \dots, f_t] \in \mathbb{R}^{T \times H \times W \times 3}$ denotes a video clip consisting of the T consecutive frames from time $t - T + 1$ to t , Z_t is all hidden vectors in the final transformer layer of VLM, n denotes the sequence length of output, and z_i represents the hidden representation of the i -th token. D_s is the token size. The VLM produces nearly identical textual descriptions across adjacent frames. Therefore, we use the semantic embedding extracted from the first frame f_1 of each video clip as the semantic representation for the entire segment. In addition, VLMs are typically built upon autoregressive language modeling, where the generation of the current token depends on all previously generated tokens. We use the final token z_n to represent the entire sequence. In this way, we obtain an extremely compact hidden-state representation H_t of the video clip, which helps alleviate computational and memory bottlenecks.

$$H_t = z_n \in \mathbb{R}^{D_s} \quad (2)$$

3.2. Multi-source feature encoding

We encode scene-related and instance-related information separately through a scene-level encoder and an instance-level encoder, respectively.

The scene-level encoder processes both the consecutive RGB frames of a video clip and the corresponding scene-level semantic embedding. We employ a Video Swin Transformer (VST) to extract four-scale feature maps from the RGB sequence, and integrate a Feature Pyramid Network (FPN) to fuse the temporal cues with these multi-level spatial features, yielding the final feature embedding of the video scene. This design preserves fine-grained details while enhancing the perception of large structural patterns, and has been shown to be effective in several prior studies [9,24]. This procedure is represented as

$$S = \text{FPN}(\text{VST}(F)) \quad (3)$$

where $S \in \mathbb{R}^{h \times w \times D_e}$ is the encoded video clip embedding. D_e is the channel dimension of the last scale feature map.

In addition, the corresponding scene-level semantic embeddings are processed by a scene semantic projection layer implemented based on an MLP. This layer maps the VLM-derived semantic vector H into a scene-level semantic representation $V^s \in \mathbb{R}^{m \times D_e}$ composed of m fixed semantic slots.

$$V^s = \text{MLP}(H) \quad (4)$$

The object detection boxes and instance-level semantics corresponding to each video clip are encoded by the instance-level encoder. The spatial coordinates of each detection box are first processed using a position encoder adapted from SAM [46], which maps the top-left and bottom-right corner points into a high-dimensional object position space by fourier positional encoding. The resulting position embeddings are then passed through a GRU module to aggregate temporal information for each instance, producing motion-aware tokens for the N traffic participants present in the video clip. Finally, a learnable CLS token is appended to the sequence of instance features for subsequent processing in the output layer.

$$O_i^t = \text{GRU}(\text{PE}(o_i^t)) = [e_i^t; e_i^{br}],$$

$$O_i = [e_i; e_i^t; e_i^{br}] \quad (5)$$

where $o_i^t \in B$ is the bounding box of object i in frame t , $\text{PE}(\cdot)$ is the fourier positional encoding, e_i^t and e_i^{br} the tokens obtained from encoding the top-left and bottom-right corners, respectively, e_i is a learnable CLS token used to aggregate the coordinate information of bounding boxes, forming a complete position representation of the object i . $O_i \in \mathbb{R}^{3 \times D_e}$ is the motion feature embeddings of object i . It is important to note that objects do not necessarily appear in every frame of the clip. To ensure the reliability of instance-level temporal features, we process objects based on temporal completeness. Objects that appear in fewer frames are discarded, while those exceeding a certain threshold are retained and completed using interpolation methods.

Similarly, the corresponding instance-level semantic embeddings are processed by an instance semantic projection layer implemented as an MLP. This layer maps the VLM-derived semantic vector H into an instance-level semantic representation $V^o \in \mathbb{R}^{m \times D_e}$ composed of m fixed semantic slots.

$$V^o = \text{MLP}(H) \quad (6)$$

In MASC, the instance and scene semantic projection layers share parameters. Since scene and object features in traffic videos are highly correlated and jointly contribute to anomaly reasoning, separating their semantic spaces would introduce undesired drift between branches. A shared MLP enforces a unified semantic space, stabilizes cross-attention fusion, and improves overall performance.

3.3. Semantic enhanced modeling

Although motion cues from the scene and instance describe temporal dynamics such as ego-motion, pedestrian crossing, or abnormal vehicle trajectories, they lack explicit contextual attributes, including object categories, functional roles, and interaction intent. To achieve robust contextual reasoning in complex traffic environments, MASC integrates both motion-aware representations and VLM-derived semantic priors into a unified semantic-augmented modeling framework. This component consists of two key modules, Language-informed Scene Modeling and Language-informed Instance Modeling, which respectively enhance the scene and instance branches of the framework.

The goal of the scene-level semantic modeling is to enrich scene visual features with semantic priors and object-level attributes. We begin by downsampling the scene feature map S extracted from the encoder to enlarge its receptive field, enabling it to more effectively retrieve high-level contextual cues from the coarse-grained semantic tokens V^s . The downsampled features are queried and fuse the scene semantics through a cross-attention layer, after which the features are upsampled back to their original spatial resolution to obtain the enhanced scene representation.

$$S' = \text{DownConv}_k(S),$$

$$S'' = \text{Cross-attention}(S' + pe, V^s, V^s), \quad (7)$$

$$S''' = \text{UpConv}_k(S'')$$

where k represents the sampling rates of the downsampling and upsampling processes, the size of the downsampled feature map is $\frac{h}{k} \times \frac{w}{k} \times D_e$, pe denotes a learnable position embedding.

Next, this enhanced scene representation is fused with aggregated object features $\tilde{O}$ that contain detailed appearance and motion cues. These object features are produced by the Language-informed Instance Modeling module, which will be described in the following. Such cross-modal fusion highlights visually and semantically important regions within the scene, allowing the model to jointly reason over static background context and the dynamic behavior of traffic participants.

$$\tilde{S} = \text{Cross-attention}(S''' + S, \tilde{O}, \tilde{O}) \quad (8)$$

The Language-informed Instance modeling module enhances instance representations by jointly integrating semantic, visual, motion, and relational information. Motion-aware tokens produced by the instance-level encoder first query their corresponding semantic slots through a cross-attention layer, injecting semantic attributes derived from the VLM, such as object category, behavioral pattern, and functional role in the scene. Next, a self-attention layer is applied with each object token serving as an anchor to model interactions among all objects. This captures inter-object dependencies, including yielding, approaching, crossing, and potential collision risks. Finally, the updated object tokens perform a second cross-attention operation over the visual feature map, injecting localized appearance information based on their spatial positions.

$$O' = \text{Cross-attention}(O, V^o, V^o),$$

$$O'' = \text{Self-attention}(O', O', O'), \quad (9)$$

$$\tilde{O} = \text{Cross-attention}(O'', S'', S'')$$

where $O = \{O_i\}_{i=1}^N \in \mathbb{R}^{N \times 3 \times D_e}$ is motion-aware tokens for the N traffic participants present in the scene. Through this three-stage enhancement, the resulting object representation encodes rich multimodal attributes, including motion (trajectory features), appearance (visual cues), and semantics (category and behavior). This enables fine-grained modeling of traffic participants and supports more discriminative anomaly reasoning.

3.4. Relation reasoning

While the previous two modules enrich scene and object representations with visual, motion, and semantic cues, traffic anomalies often arise not from isolated entities but from the relationships among them. Interactions such as a vehicle approaching a pedestrian, a vehicle drifting out of its lane, or a vehicle colliding with a roadside barrier are fundamentally relational. Therefore, modeling a unified set of scene-instance and instance-instance relationships is crucial for capturing the complex relational cues of traffic anomalies.

To integrate all relational cues into a coherent semantic space, we follow PromptTAD on relation-wise aggregation modeling and construct a unified relation reasoning module built around a set of learnable query tokens. These learnable queries act as relation reference anchors that progressively aggregate multi-level information from both the Language-informed instance and Language-informed Scene branches. Specifically, the learnable queries interact with instance-level and scene-level features through two cross-transformer blocks, which are composed of layer normalization, cross-attention layers, and feedforward network layers [47]. In the first block, the queries attend to the instance representations to capture object-object interactions, such as spatial proximity, behavioral influence, and potential conflict patterns. In the second block, the updated queries attend to the enhanced scene representation to incorporate broader scene-object and scene-scene relationships, including structural layout and environmental context.

$$Q' = \text{Cross-transformer}(Q, \tilde{O}_{cls}, \tilde{O}_{cls}),$$

$$\tilde{Q} = \text{Cross-transformer}(Q', \tilde{S}, \tilde{S}) \quad (10)$$

where $Q \in \mathbb{R}^{g \times D_e}$ denotes a set of learnable query tokens and g is the number of tokens. $\tilde{O}_{cls}$ is the CLS token of object representation, which contains complete positional embeddings. Through this sequential cross-attention over object features and scene features, the queries accumulate a unified set of semantically aligned relational cues, enabling the model to reason over comprehensive interaction patterns essential for anomaly detection.

3.5. Anomaly decoding

MASC adopts a dual-branch decoding architecture to construct a joint optimization objective consisting of scene-level and instance-level supervision. While the scene branch models global anomalies, the instance-level loss is introduced as a structural constraint, providing clearer optimization guidance within the overall loss space. This collaborative loss design helps mitigate optimization ambiguity caused by complex dynamic backgrounds, thereby promoting more stable and accurate convergence toward anomaly targets [48].

$$\mathcal{L} = \mathcal{L}_{scene} + \lambda \mathcal{L}_{ins} \quad (11)$$

where $\mathcal{L}_{scene}$, $\mathcal{L}_{ins}$ denotes the scene-level loss and instance-level loss respectively. λ is the weight of the instance-level loss. In the following, we present the decoding process and the corresponding loss formulation for each branch in detail.

The scene-level decoder serves as the primary prediction branch. Traffic anomalies often manifest as abnormal patterns within the global semantic relation of the scene, such as unusual vehicle-vehicle interactions, deviations from road topology, or conflicts between dynamic objects and static infrastructure. After unified relational reasoning aggregates all scene-object and object-object relations into a coherent relational embedding, the scene-level decoder processes this relational embedding using a lightweight regression head implemented based on an MLP.

$$\hat{y} = \text{MLP}(\tilde{Q} + S) \quad (12)$$

where $\hat{y}$ is a scene-level anomaly score for a video clip. For supervision, we compute the binary cross-entropy loss between the predicted score $\hat{y}$ and the ground-truth label y .

$$\mathcal{L}_{scene} = -\alpha y \log(\hat{y}) - (1 - \alpha)(1 - y) \log(1 - \hat{y}) \quad (13)$$

where α is a scaling factor used to balance anomalous and non-anomalous samples.

The instance-level branch provides complementary object-level supervision, improving the model's sensitivity to fine-grained abnormal behaviors such as sudden lane changes, abrupt braking, or pedestrian intrusions. For each object i in a video clip, the instance decoder first queries the unified relational encoding through a cross-transformer using its enhanced object representation, thereby retrieving the scene relations associated with that object. It then applies an MLP to predict the anomaly scores for all N objects in the scene $\hat{X} = \{\hat{x}_i\}_{i=1}^N$.

$$\hat{O} = \text{Cross-transformer}(\tilde{O}, \tilde{Q}, \tilde{Q}),$$

$$\hat{X} = \text{MLP}(\hat{O}) \quad (14)$$

To obtain supervisory signals, we follow PromptTAD and use Hungarian matching to align the detection boxes provided by YOLOv9 with ground-truth anomalous objects, thereby producing binary labels, as shown in Fig. 2. The object-level loss is then defined as:

$$\mathcal{L}_{ins} = -\frac{1}{N} \sum_{i=1}^N (\alpha x_i \log(\hat{x}_i) - (1 - \alpha)(1 - x_i) \log(1 - \hat{x}_i)) \quad (15)$$

Here, $\hat{x}_i$ denotes the predicted anomaly score of object i . x_i denotes the ground true label of object i . Specifically, if the detection box of object i provided by YOLO matches any ground-truth anomalous object through Hungarian assignment, its label is set to 1; otherwise, it is assigned 0. This auxiliary branch encourages the model to understand object-specific anomalies and strengthens the learned relational semantics.

Table 1

Traffic anomaly categories in the DoTA dataset.

Label	Anomaly Category
ST	Collision with another vehicle that starts, stops, or is stationary
AH	Collision with another vehicle moving ahead or waiting
LA	Collision with another vehicle moving laterally in same direction
OC	Collision with another oncoming vehicles
TC	Collision with another vehicle that turns into or crosses a road
VP	Collision between vehicle and pedestrian
VO	Collision with an obstacle in the roadway
OO	Out-of-control and leaving the roadway to the left or right
UK	Unknown

4. Experiments

4.1. Datasets and metrics

We evaluate our method on the widely used traffic anomaly detection benchmarks, DoTA [31] and DADA-2000 [49]. The DoTA dataset contains a total of 4,677 video clips captured from dashcam perspectives, covering diverse locations, weather conditions, and lighting environments. Each video contains a clearly defined anomalous. All videos are in 1280×720 resolution and are precisely annotated with the start and end times of each anomaly, the trajectory bounding boxes of the involved objects, and one of nine predefined anomaly categories (see Table 1). The dataset is further divided into multiple subsets based on whether the anomaly involves the ego-vehicle and on the specific anomaly categories. The DADA-2000 dataset consists of 2000 dashcam videos with a resolution of 1584×660 pixels, each annotated with the spatio-temporal extent of the anomaly and one of 54 anomaly categories. The dataset is further divided into two subsets: ego-involved anomalies and non-ego-involved anomalies.

We adopt the Receiver Operating Characteristic (ROC) curve and its Area Under the Curve (AUC) as the primary metric to evaluate the performance of the proposed method. The AUC is computed as the area under the standard frame-level ROC curve, which characterizes the relationship between the true positive rate (TPR) and the false positive rate (FPR) across different thresholds. A higher AUC indicates stronger discriminative capability in distinguishing anomalous frames from normal ones. In addition, we further provide qualitative evaluations through anomaly score curves and attention visualizations, offering a more intuitive assessment of the model's behavior.

4.2. Implementation details

We train and evaluate the proposed method on an Ubuntu 22.04 environment equipped with four NVIDIA V100 GPUs. Following prior work [9], the training setup employs the SGD optimizer with an initial learning rate of 0.002, a momentum of 0.9, and a weight decay of 0.00001. The scaling factor for the loss is set to $\alpha = 0.7$, and the weight of the instance-level loss is $\lambda = 1$. The model is trained for 400 epochs with a batch size of 2. Each video clip consists of 5 RGB frames with a resolution of 640×480 . Based on the physical meaning of motion, objects that appear in fewer than 3 frames within the input frame sequence are discarded. The downsampling rate of scene feature is set to $k = 5$. The feature dimension D_e of the model is 256. The number of fixed semantic slots m is 8. The number of learnable query tokens g is 12. For semantic augmentation, we use Qwen2.5-vl with 3 billion parameters [50] as the VLM-semantic extractor, and the semantic vector dimension d_s is set to 2048, corresponding to the token size of the last Transformer layer in Qwen2.5-vl-3b.

4.3. Comparison with state-of-the-arts

To validate the effectiveness of the proposed method, we compare MASC with seven state-of-the-art approaches: FOL [31], MAMTCF [8],

Table 2

The AUC of different approaches on DoTA dataset.

Method	Type	Input	AUC (%)
FOL	Unsupervised	RGB + Box + Flow + Ego	73.0
MAMTCF		Box + Flow	76.6
TRN	Supervised	RGB	78.00
STFE		RGB+Box	79.30
MOVAD		RGB	82.17
TTHF		RGB	79.43
PromptTAD		RGB + Box	83.87
Our		RGB+ Box+Sem	84.26
TTHF ^o	Post-processing	RGB	84.67
PromptTAD ^o		RGB + Box	85.76
Ours^o		RGB+ Box+Sem	87.30

TRN [51], STFE [25], MOVAD [24], TTHF [11], and PromptTAD [9]. Among these, FOL and MAMTCF are unsupervised methods, while the remaining approaches are supervised.

Table 2 summarizes the performance of all comparison methods on the DoTA dataset, along with the corresponding input configurations used in each method. Methods marked with “o” adopt the post-processing techniques introduced in TTHF, including median filtering and min-max normalization. The results demonstrate that the proposed MASC achieves the best overall performance, significantly outperforming unsupervised methods due to the inherent advantages of supervised learning. Among the supervised baselines, the most relevant competitors are TTHF and PromptTAD, which share conceptual connections with our framework. Compared with TTHF, MASC obtains a +2.5% AUC improvement. TTHF integrates textual semantics through carefully designed fine-grained text prompts and performs contrastive learning in a CLIP-like framework. In contrast, our method exhibits superior performance despite using coarse and label-free VLM semantics, demonstrating that such weak textual priors can still provide meaningful semantic cues for scene understanding without relying on manually crafted labels. MASC also surpasses PromptTAD by +1.5% AUC. Considering that driving video anomalies typically arise from dynamic behaviors, we reformulate the static prompt-based mechanism in PromptTAD into a dual-branch dynamic feature interaction framework. Experimental results confirm that our design successfully leverages motion-aware representations to capture richer anomaly cues from temporal dynamics. Moreover, motion features complement the injected semantic signals, making the semantic augmentation more effective, as further validated in our ablation studies.

To further evaluate the effectiveness of the proposed framework in detecting different types of traffic anomalies, we conducted a classification analysis on the DoTA dataset. Table 3 reports the AUC performance of our method and previous approaches across all anomaly categories, whose definitions are provided in Table 1. Categories marked with an asterisk (*) correspond to non-ego anomalies, while the remaining categories represent ego anomalies. Our method achieves either the **best (Bold)** or suboptimal (Underline) performance in most anomaly categories. Compared with prior supervised approaches, MASC maintains consistently strong performance across motion related categories (excluding ST and VO), which emphasizes the effectiveness of our motion-aware and semantic-augmented framework. For static anomaly categories, our method still demonstrates a substantial improvement in the ST category. Performance on the VO category is comparatively weaker, primarily because the static vehicles in the ST category typically appear as large objects, the VO category consists of small static obstacles. Due to the model's focus on dynamic cues and the inherent limitations of the object detector, these small objects are more likely to be missed by the detector and subsequently filtered out by our pipeline.

Additionally, Table 4 presents a comparison between the proposed method and previous state-of-the-art approaches on the DADA dataset. Although our model is not trained on DADA dataset, it achieves the best performance for both ego anomalies and non-ego anomalies,

Table 3

The AUC (%) of different approaches for each anomaly category on the DoTA dataset.

	ST	AH	LA	OC	TC	VP	VO	OO	UK
FOL	73.3	81.2	74.0	73.4	75.1	70.1	-	-	-
MAMTCF	64.0	78.7	82.5	71.4	77.6	66.5	76.8	79.2	71.8
STFE	75.2	84.5	72.1	77.3	72.8	71.9	-	-	-
TTHF	84.3	86.3	84.8	83.2	85.3	73.6	81.6	86.2	65.2
MOVAD	86.6	86.3	84.9	83.7	85.5	81.6	77.4	87.9	73.8
PromptTAD	86.2	89.1	87.6	85.7	87.9	82.2	84.4	86.8	73.6
Ours	88.6	88.1	87.4	85.5	89.2	82.7	82.5	88.1	73.9
	ST*	AH*	LA*	OC*	TC*	VP*	VO*	OO*	UK*
FOL	77.5	69.8	68.1	76.7	73.9	71.2	65.2	69.6	-
MAMTCF	78.8	74.9	76.1	82.0	76.8	86.7	77.2	76.9	76.7
STFE	80.6	65.6	69.9	76.5	74.2	75.6	70.5	-	-
TTHF	68.5	69.4	69.6	74.3	75.3	72.8	77.3	71.6	63.7
MOVAD	72.2	74.0	74.8	80.2	79.6	76.8	82.8	78.3	72.9
PromptTAD	70.9	74.3	78.4	82.9	81.5	78.3	83.5	79.1	77.5
Ours	73.3	75.8	77.9	85.2	82.5	81.9	80.1	79.3	73.9

Table 4

The AUC (%) of different approaches on DADA dataset.

Method	Ego	Non-Ego	Both
AnoPred	55.7	56.9	56.1
FOL-STD	71.3	57.1	66.6
DMMNet	73.0	56.3	67.5
SSC-TAD	67.6	58.7	71.7
MAMTCF	79.7	62.1	70.2
TTHF	80.9	64.0	71.7
PromptTAD	79.7	70.4	74.6
Ours	81.4	70.1	75.6

Table 5

The AUC for variants of our method on DoTA dataset.

Ablation	Sub clip	Obj-seq	Sem2obj	Sem2img	AUC (%)
0	\	\	\	\	55.23
1	✓	×	×	×	8 2.95
2	✓	×	✓	×	8 3.17
3	×	×	×	×	82.58
4	×	×	×	✓	82.95
5	×	×	✓	×	8 3.27
6	×	✓	×	×	83.71
7	×	✓	✓	×	8 4.06
8	×	✓	✓	✓	84.26

demonstrating strong generalization capability. This robustness can be attributed to two key factors. First, the vision-language model provides a certain degree of zero-shot semantic understanding, enabling it to supply stable semantic vectors even without explicit supervision. Second, the enhanced motion representations exhibit strong resilience to domain shifts and various forms of scene noise, making the model more sensitive to abnormal motion patterns across different datasets.

4.4. Ablation studies

In this section, we conduct ablation experiments on the DoTA dataset to evaluate the contribution of the key components in MASC. These components include the instance-level temporal features (Obj-Seq), semantic injection into object representations (Sem2obj), and semantic injection into scene representations (Sem2img). The results are summarized in Table 5.

We first re-train the original PromptTAD framework with original training settings under our hardware (Ablation 1). Compared with re-trained PromptTAD, our full model (Ablation 8) achieves a substantially higher AUC, demonstrating the overall effectiveness of the proposed framework. We further verify each component through Ablation 3 to Ablation 8. Using Ablation 3 as the comparison baseline, the results

Table 6

Accident keyword matching system.

	ACCIDENT (=1)	NO ACCIDENT (=0)
Keywords	crash / collided / collision / rear-ended / sideswiped / struck / impact / smashed / rollover / overturned / etc.	no accident / no crash / didn't hit / avoided collision / almost hit / near-miss / managed to stop / etc.
Examples	"The ego vehicle crashed into the car ahead, causing a rear-end collision." "A truck sideswiped the ego vehicle while changing lanes." "The vehicle ahead hit the guardrail on the left side of the road"	"The vehicle stopped at the intersection and waited for the pedestrian ahead to pass" "The traffic was heavy but there was no collision or crash." "The car almost hit the cyclist but managed to stop in time."

clearly indicate that each individual module contributes positively to performance improvement.

In addition, we conduct a separate discussion on the Sub-clip mechanism, which is a key component of the PromptTAD framework. PromptTAD extracts temporal features from multiple sub-segments within a video clip, enabling it to capture continuous scene variations and obtain more stable anomaly signals over a longer temporal window ($T = 8$). In contrast, MASC performs anomaly reasoning using independent video clips with a shorter temporal window ($T=5$). Although removing sub-clips introduces a slight performance drop (Ablation 1 vs Ablation 3), this gap is largely compensated by injecting object-level semantics (Ablation 2 vs. Ablation 5). This is because shorter video clips produce more stable and less varied visual patterns, making them easier to align with semantic representations, thereby reducing reliance on long-range sub-clip modeling.

A comparison between Ablation 2 and Ablation 7 further demonstrates that the alignment between motion features and semantic features is particularly effective. Motion cues naturally reflect the semantic attributes of traffic participants making them inherently easier to map into a unified semantic space, for example, "a vehicle moving from right to left" or "a pedestrian approaching from the left". Therefore, a motion-aware framework provides a more effective solution.

The ablation results about VLM-semantic (Ablation 6 vs Ablation 7 vs Ablation 8) demonstrate that these semantic cues indeed contribute useful information to the model. However, it remains unclear whether the VLM inadvertently supplies direct anomaly-related labels, which could provide the model with an unintended shortcut. This potential issue warrants further investigation. Although the VLM occasionally produces words such as "accident" or "crash" in its free-form descriptions, we find these expressions to be rare, inconsistent, and semantically unreliable. Such terms do not reliably correspond to actual accident scenes. In many non-accident cases, they incorrectly appear as hallucinated descriptions, while many true accident cases lack any

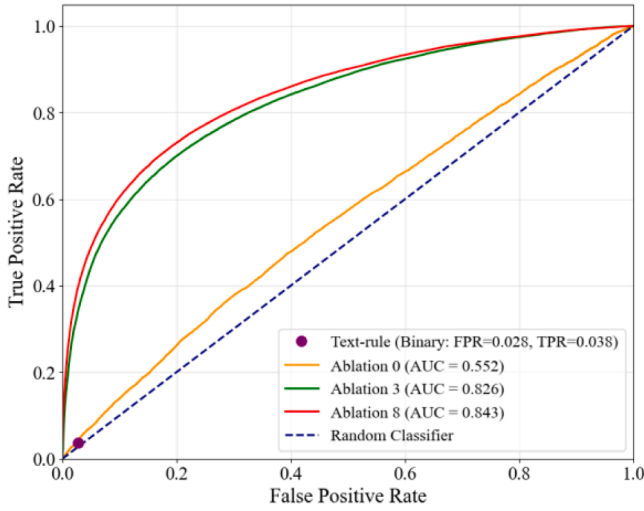


Fig. 4. Comparison of ROC curve across different methods.

Table 7

Performance under different VLM selections and prompt templates.

VLM	Prompt Description	AUC (%)
Qwen2.5-VL-3B	"You are to observe driving images as an experienced driver. Please briefly describe the traffic scene" (Scene prompt)	83.89
Qwen2.5-VL-3B	"You are to observe driving images as an experienced driver. Is there any accident risk in this driving scenario? If so, indicate the location of risk." (Risk prompt)	83.89
Qwen2.5-VL-7B	"You are to observe driving images as an experienced driver. Please briefly describe the positional relationship between all traffic participants within the detection boxes"	84.09
InternVL3-4B	"You are to observe driving images as an experienced driver. Please briefly describe the positional relationship between all traffic participants within the detection boxes"	84.21
Qwen2.5-VL-3B	"You are to observe driving images as an experienced driver. Please briefly describe the positional relationship between all traffic participants within the detection boxes" (Relation prompt)	84.26

explicit accident-related wording. To quantify this phenomenon, we further analyze the raw text generated by VLM using a regular-expression based text-rule system that searches for common accident-related terms and assigns a binary prediction (as shown in Table 6). Fig. 4 shows that the resulting TPR and FPR are extremely low, indicating that explicit accident-related words rarely appear and are almost unrelated to the ground-truth labels. These results indicate that direct accident expressions in the text are neither stable nor discriminative signals.

In addition, an extremely lightweight MLP classifier trained on the semantic embeddings from VLM achieves an AUC of only 0.55 (Ablation 0). This suggests that the semantic embeddings do not contain direct or linearly separable accident-related signals.

Overall, these findings confirm that the VLM does not provide accident labels or explicit accident cues to the model. Instead, the VLM contributes scene-level semantic priors, such as object roles, spatial context, and environmental descriptions, rather than accident-specific information. This ensures that the use of VLM-semantics does not introduce label leakage or unfair advantages, and the improvements primarily stem from enhanced contextual understanding rather than any form of shortcut.

4.5. Sensitivity analysis

To analyze prompt sensitivity, we evaluate three types of prompts, including scene prompt, risk prompt, and relation prompt (Table 7). All

semantic prompts consistently outperform the ablation model without semantics (i.e., Ablation 6 with an AUC of 83.71% in Table 5), demonstrating that the semantic prior provides effective information. The relation prompt achieves the best performance, suggesting that spatial relationship information between traffic participants is more beneficial for anomaly detection. Meanwhile, the overall performance variation across prompts remains small, indicating that the model is robust to prompt formulation. We further investigate the impact of different language backbones by replacing Qwen2.5-vl-3B with InternVL3-4B and Qwen2.5-vl-7B. The performance differences are marginal, showing that the proposed framework is not sensitive to the choice or scale of the language model. Since language embeddings are frozen and serve as high-level semantic anchors, the performance improvement is primarily due to the structured motion-semantic modeling design, rather than an increase in the capacity of the language model.

4.6. Visualizations

To more intuitively demonstrate the practical effects of the proposed framework, we have visualized the complete detection process and the evolution of the model's internal attention. Fig. 5 includes several representative visualization examples, including ego vehicle loss of control, non-ego vehicle loss of control, ego collisions, and non-ego collisions. It presents the attention weight evolution across pre-collision frames, collision frames, and post-collision frames. The first row shows the raw RGB frames, where red bounding boxes denote the ground-truth anomalous regions. The second row shows the attention maps of the baseline (i.e., Ablation 1 in Table 5). The third row presents the attention maps generated by the ablation model without semantic information (i.e., Ablation 6 in Table 5). The fourth row is the result of the proposed model. The fifth row illustrates the anomaly score curves over the full video sequence. Fig. 5(a) features an ego-vehicle loss of control scenario with no visible traffic participants. As the ego vehicle deviates from the lane, MASC establishes stable relational cues between the vehicle and off-road scene elements, such as roadside areas and surrounding trees. Fig. 5(b) illustrates an ego collision case. The proposed model consistently attends to the moving vehicle in the lower-left region, indicating that MASC allocates stronger attention to dynamic agents, which is consistent with our motion-aware design principles. This demonstrates that the injected motion and semantic cues enable the model to prioritize behaviorally informative regions. In Fig. 5(c) and (d), which depict non-ego anomalies, MASC consistently maintains strong attention on the anomalous area, validating the model's robustness to diverse anomaly patterns. In contrast, the baseline and the ablation model without semantic information exhibit weaker and less consistent attention to the critical area.

The anomaly score curves shown in Fig. 5 reveals an important observation: the proposed model exhibits earlier anomaly awareness. For example, at frame 40 of case (b) and frame 51 of case (d), the model raises anomaly scores before the actual accident occurs, indicating its ability to anticipate potential collision risks. To better understand this behavior, we examined the VLM text generated at frame 51 of case (d): "Here's a detailed description of the positional relationships between the traffic participants in the image: 1. White Car: Positioned on the left side of the image, slightly ahead of the ego vehicle. The car appears to be moving forward. 2. Yellow Taxi: Positioned on the far-left edge of the image, partially visible. The taxi is moving in the same direction as the ego vehicle." The text provides explicit semantic cues about the relative positions and possible motion intentions of the involved vehicles. When combined with the motion features modeled in our framework, these semantics make it easier to infer potential interaction risks. Fig. 6 presents the anomaly rate per frame during anomalous duration, indicating that our model delivers earlier responses at the early stage of anomalies. This early risk-awareness behavior suggests that the synergy between motion cues and semantic priors in MASC may offer promising insights for future

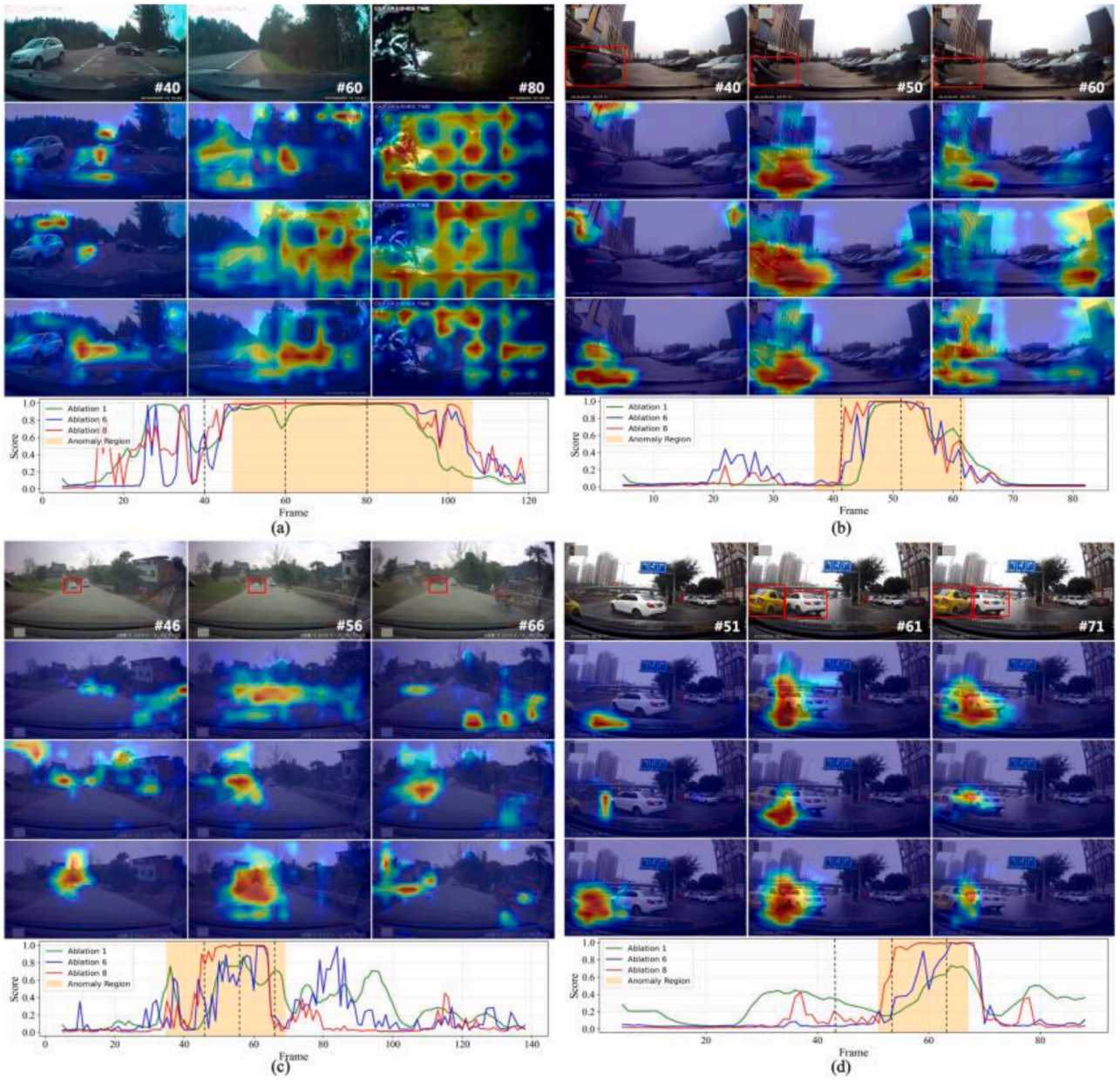


Fig. 5. The visualization of anomaly score curves and attention maps of different variants.

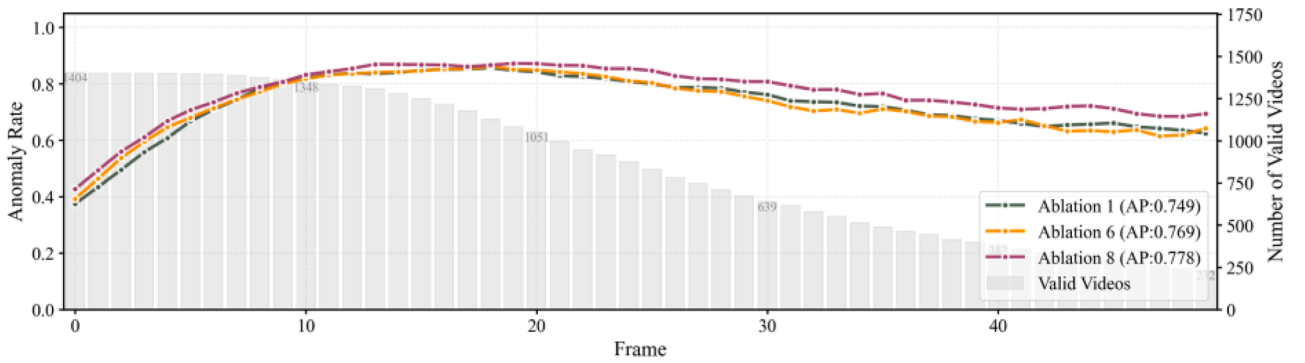


Fig. 6. Anomaly rate per frame during anomalous duration. The horizontal axis denotes the frame index counted from the begin of the anomaly. The bar chart illustrates the number of valid video samples available for each frame index. Since only a few videos contain anomalous segments lasting longer than 50 seconds, the statistics are truncated at 50 seconds. The anomaly fixed threshold is set to 0.5. The value of AP refers to the area under the Precision-Recall (PR) curve, where the PR curve reflects the model's ability to detect positive (anomalous) instances.



Fig. 7. Model attention changes for traffic participants.

research on collision prediction and early-driving risk assessment.

In the relational reasoning module, each relational query is designed to learn potential abnormal relationships within the scene. Specifically, the query first embeds objects information to identify key targets that may involve abnormal interactions, and then interacts with frame features to highlight scene regions most relevant to the object interactions encoded in the query. Fig. 7 illustrates the temporal evolution of attention over traffic participants within the relational reasoning module. It can be observed that the model progressively focuses on the key interacting objects involved in the anomaly and maintains stable attention throughout the anomaly development process. Meanwhile, the model also attends to the scene regions most relevant to these object interactions, as shown in Fig. 5. In addition, Fig. 5 compares the early-stage attention maps between the proposed model and the ablation model without semantic information. The results show that, with the introduction of semantic alignment, the model can more accurately focus on potential anomalous regions before the anomaly fully manifests, thereby demonstrating the importance of semantic information for early detection. Overall, the proposed model detects anomalies through semantic relational reasoning rather than relying solely on abrupt pixel changes. The semantic interaction is the key factor enabling the model's early detection capability.

5. Conclusion

In this work, we proposed MASC, a cross-modal semantic-enhanced framework for traffic anomaly detection that unifies multi-level motion modeling with semantic reasoning. Unlike traditional approaches that rely primarily on pixel-level dynamics, MASC elevates video representations to a semantic space by directly injecting raw VLM hidden-state embeddings into motion features. Through semantic-aligned motion modeling and a unified relational reasoning module, the framework captures object-object and object-scene interactions, enabling context-aware detection at both scene and instance levels. On DoTA and DADA dataset, MASC achieves competitive performance with over 1.0% AUC improvement. Extensive comparative experiments have validated that the proposed framework exhibits strong robustness in terms of visual language model selection, prompt paradigms, and scene categories, while ablation studies confirm that aligning motion perception with semantic priors effectively enhances contextual reasoning capabilities and early anomaly perception.

Despite these advantages, the framework still faces challenges in computational efficiency and deployment. The VLM-based feature

extraction with 3B parameters runs at 5.44 FPS, while the anomaly detection module with 126M parameters achieves 16.2 FPS, which does not fully satisfy strict real-time requirements. In addition, the large parameter scale of the 3B VLM limits its applicability in edge scenarios. Future work will therefore focus on distilling large-model semantic knowledge into lightweight, task-specific encoders to build a more compact, efficient, and practical end-to-end framework.

CRedit authorship contribution statement

Hao Tong: Writing – review & editing, Writing – original draft, Visualization, Methodology, Data curation, Conceptualization. **Chengcheng Xu:** Validation, Supervision, Project administration, Funding acquisition, Conceptualization. **Dejie Yang:** Software, Methodology. **Qi Ai:** Writing – review & editing, Data curation. **Haibo Chen:** Writing – review & editing, Supervision. **Haiguang Lai:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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Data availability

Data will be made available on request.

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